

A Packing of 64 Lines in $\mathbb{R}^{13}$

Limin Ge*

October 8, 2026

Abstract

We give 64 1-dimensional subspaces of $\mathbb{R}^{13}$ whose smallest squared chordal distance is at least 0.906293. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 0.905318, obtained by deleting 1 of the 65 lines of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 64 subspaces of dimension 1 in $\mathbb{R}^{13}$ whose pairwise squared chordal distances $D(a, b) = 1 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 0.906293\dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 13$, $n = 1$. The cell $N = 64$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 0.905318, obtained by deleting 1 of the 65 lines of the packing found by this search. The Rankin simplex bound for this cell is $1(13 - 1)/13 \cdot 64/63 = 0.937729$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m13n1N64.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 70 seconds on one laptop, with about ten such searches running at once.

A Exact value

```
numerator:
302097752527338792774504204448064938477392188503
denominator:
333333333333351337191823992064640902847704214770
```

B Coordinates

64 blocks of 13 rows with 1 integers each; divide by 10^{12} . File: `grassmann/packings/m13n1N64.txt`.

294513571522	-116646900031	333927834310	-73715824796
-146842871179	-163785267506	27797377517	42769951193
-366546568701	48354694415	-212050586567	-408176489198
-266440092016	-155343363799	105704188561	-71279519144
-62499393211	383649740472	-502849924790	-139371010381
-532003397380	175078807820	-36425411676	15013881487
32919873544	647584521720	216898196568	46755000706
233986742826	-150678986713	277191491280	-86815946117
-131445619310	-2071058650	-376921818680	-27209426925
-222056811586	-147681425053	-60370076814	108229387020
-224579466453	-369856534562	121797516501	820626425858
81809328830	411555871148	605876284433	-315872447757
-468901944336	418260454548	-78568158972	163298611788
534597576460	420760575	81423544235	229836590465
-297068154461	-343065269142	-62853857515	461799870697
-195089019629	357837986808	-257629868864	42928209076
-239483350678	-50161648682	-318722108501	-85460955596
279454576937	161745687421	274513596468	64891545629
-137201631177	358878153374	-306699380665	211772576594
-515908018848	251580864714	329528966758	-401359131440
45929330591	174830514343	280454789332	-443999296472
3174462625	-409063344687	500146214807	-54138566585
-87241517828	504939965841	-148225650673	211873038517
-136347174288	-116673497616	-238175662846	378448862843
-296872158317	-153926656417	-166279932686	184110028971
225852643467	461010816082	-156270216788	-200131420120
-94483369526	-186016482579	-107079900821	220550668282
113207819447	30748319157	363336471316	-63026745744
-163430411557	-140922451845	-131742721701	172958363997
494462116755	78407193632	229478692821	48907108512
-391566923088	165047926123	275543214282	806726981744
226085336444	264805102798	221170231561	-145936611519
-164999898795	262457359660	11740340276	292796305490
-93468799185	316330326381	-337550211014	135475962172
-363027775797	174327750844	228989641415	190094686960
-31712231669	423976884568	-321382049850	-62053675799
566932170651	-120857040123	219216913589	-157966654181
-106604461877	307104692013	6323705923	-304540994264
-32901693848	298858161612	-126673133184	362307906104
218987175767	-50619629139	85240763026	-123949228488
-9233141269	-148568069164	-640309767281	291183144509
46554488110	209752892929	267009033558	170260598996
-128906470071	146402458338	-61892670286	45223483245
-184396193348	76696626466	-174946874445	-497315969268
20351089885	-369174826591	44297974756	-98294880597
36532394132	-115373236973	-76456329914	423575986576
-422621517760	588276582089	532655965084	7929476370
-326572119529	474637886254	-264889863083	261170064510
-620643519367	261280157506	26384743337	-325979764117
329846201992	633307131303	-37649468918	-184026452016
284133388158	116725425363	-72119652468	-240876740731
193487609077	-167038653009	424767915274	243503579428
102045538404	284237968446	473986053277	420478339094
-103642840105	155832035274	-435146717540	-81364810073
114797728994	-136324191756	-45911687359	95048423201
-687985479553	-109312090862	-178434364137	-444392509248
-229129377828	131244104898	156393080303	111896796500
-296868835667	66123485531	-49968274490	66251756478
-389884867512	281079069565	-89626763893	-443569794640
-233985588744	-166240365117	-287773933516	270251623630
-164160721551	-323104463489	-205399816883	-366155325618
126463873869	302739332608	249576178931	220921067957
-23287043852	579131253996	559859719391	-152207423424
313862855136	-129140569149	-122016679389	-67498436354
54382496818	1587392295	-157622943093	-7527955810
48522283785	52383439036	-515000981643	167808834318
373559234535	-394904424323	-322643086404	-152359303113
-291380515795	-72803942611	152908032025	-99531817436
65407164541	-271557392517	132152281970	-555254201751
-286543106189	-238529415133	-297202370327	-295432928778

-55891284887	-120494830157	41566006154	-693592792693
263023710660	307020623254	-163505191425	114286789610
-632362152281	-303498219061	-202671964952	403556618226
-178320539298	120948858560	-426572648345	189425353976
-59456095484	107064438055	-153082051793	393016502423
176066068398	139113181081	264602259392	141359239173
-61125715683	-726523827743	111332990570	-114786442421
-136985415693	-3299732655	-498405526835	-246844158658
494235753	356073139691	127998750575	-74446224123
397490424241	191081855484	-501270814579	-29400210888
-217352833392	217093549438	23698937443	163997127687
423883736508	206542460035	-553128227900	103418586365
-198240930990	586477566864	-179281859445	99469914461
287762945115	259683370493	-61556624529	-70716141971
400045274922	-106366712274	-82420705797	-461629002726
115087121789	-31533286309	-323869926765	222780017312
92405062420	-314746714310	-28643952550	-77966924767
327418208688	-266637827629	-650451537936	235688600294
-425970021299	287151377505	-8480042187	27087012338
-173917331263	-200777432673	-74130266952	-148023729367
-503848129042	174016997313	-290030614638	-404593799743
332467688403	450358711227	-156802103630	-429511074081
-55735195158	69546208329	-87703332678	-323600946841
-144890245349	150983780495	322128804949	-334467199559
212444096712	10438732131	-510693021802	270810296750
-648631334022	-44366957922	104311003281	-217144222730
-175483530187	-275027950249	134772088403	-139940295780
267244622702	-192833223169	81092080605	97878627665
-62567184336	-279992390140	-215987379215	-332807717888
-12591840928	56464845463	-186299058680	-78387783212
-69341900464	-295462469348	-33382282104	-194253882410
65610213620	364763981475	-113243812493	-369442454902
511007973190	-387367294065	666557681843	51675587982
181499557611	-450170594222	-112200350012	-262564453347
7616173511	405145922032	217166662460	87008722598
400903484119	-255770314030	-21438228169	-207468006856
-58386507734	236370801752	46597529614	-609014899263
-68136932773	-144015387675	-196113153317	-373408431459
13964737516	38467967714	450998818429	570801370695
119571895761	532081877901	461140374237	-140208466521
377791939397	-5282413440	-142626085976	16160975232
448713434652	367260600467	-158890531464	-323159861893
384646017256	282672666554	316956185214	108487446072
-168109848651	280379408566	246665149887	40988398469
48157019096	-138614694509	281532615820	-43728871328
270468663197	-6377284638	94774784529	340116462499
-376975928537	310735299214	-432553015841	-233426788598
163425897238	-34909538363	223666817363	126610578947
-252609293067	503873882959	105132160610	530927861521
-195287438132	555507331250	161322907338	173598346038
287795212414	246114275100	235270444825	-190404376085
-7985420540	161183678420	-382471512236	10891515906
-379862717138	-329399540717	-501761043286	-219610039243
-345282163281	269858465404	110141206422	-219637867345
259991718922	-134050247763	-312172718	-106380894013
269875081017	193943158383	-38503762289	588016104417
-324968158271	-106365526447	350456666294	93337165927
252272322450	73679027737	-115103778040	133933846232
-389137096712	-289058829639	-325416263799	429433055884
-254420154933	111996994660	-62087776909	-473055070798
-24151458444	-82730776257	328321216549	58926246945
-221500643906	-521920196791	391594432956	-285488303377
596546640511	-11433342734	295859674103	-152258705289
442508848857	1934103440	-57170776594	-57703038256
-19777060512	-68316921443	-59069294743	-262613493563
112312524861	-454286731947	-367503130011	-146376381353
374876276008	-177683708654	142084624147	141093560111
34212172443	136823332462	681652457171	204182625483
130414580842	-584119636579	266654154972	674879590198
63896982551	-119623663277	-75807994956	-264939370651
-81952322804	296908222179	-141260256200	-438146775148
-136231893777	-21737003952	-326711805711	-222442618533
-113122038368	123094041943	-73715991050	-93267202670
422081978362	-201206320265	-17726127470	-84604936587
36705448321	-186959375554	-276619039909	-253831731464
-604389773877	-334620053758	309808239735	10702031432
64009731797	591116330282	58660992552	-22353444108
-232747586933	-192166035026	-333481560887	-31151016331
399469770724	-163228342785	343742540129	-36922862205
-263421863374	4094737930	-271582864553	232056435476
332036324309	233468442711	117638161997	-539249081303
35834552393	-370333757994	321039669545	-258789235719
136317553388	155724176767	-452755623008	161998680738
80661133205	-322815235757	49644432903	-183640840578
540796013391	98850474298	-48318477390	517839312946
112942842855	-264006345570	-13354891492	280776879062
138911868233	298991198946	176515469143	-304901097897
106279624244	-40637977259	-489309140715	-36877399372
238303739304	5865818940	26416040143	14411340541
-33436887683	30799855716	-2025851934	-290794837327
201021330440	84947351969	-384185536579	-358807870841
-170717843359	64524413433	-95741166591	98433654498
-113495985777	-302387444799	87419827547	-32703998042
289197199057	-272833681414	-61832224805	-714833413858
-431993532673	-721528311745	-442546494998	38382451071
-480195049102	-13541136489	-179910229470	40928933400
-151686549640	-140506750637	586016526647	392105197894
-70847522208	127224085698	72472702639	203174032606
177168565157	-417132304922	-162393822039	-120249110442
-357968833422	204018317215	426893830251	331018161925
62358744607	122387572547	213482767923	67556901548
-239537335854	264553008335	-120740268828	-112847836780

-100026802657	-486589025241	-48409037108	9944948378
173678231075	-129935630656	-278806258502	369650743497
390396014782	-107228061815	-180142834326	-528258917944
192371951072	31996651227	313021792978	-87974422573
-205477260620	-162908491457	-558134789164	-170803763375
293079994306	-269942770259	19544745175	-9335171285
214159698708	-131343139422	-120019885294	7931746868
-531885993398	188129130701	-282993665945	2332576792
-56684334914	210686818014	37783197272	381533222711
-88782807276	172563905141	-176354739983	54742009448
557062633272	-304671358078	68746418000	148325497588
-20595155850	256025458297	-212829348466	-136563274195
-37409521550	-136005452548	-320690767696	-464365062811
17264735393	-601496386198	138077138513	-174222217429
-306630862865	168531397451	536465287400	-108546819797
-29877330243	-162025212331	78996239153	-491100356633
-318562079253	-424802117042	49027764209	155545826116
-329254411582	-428846942299	473052584961	-211623274998
-151520870973	-338273813528	-219494752409	181109590719
344292159915	522915925419	-383483934449	49366114976
307381629216	-149928561246	-126793991184	558968414828
-598644320771	-78362051737	-4616020663	160748376728
178232073720	-291612220368	236833642070	133781212545
2977413681	256927067538	-281414004722	127346913565
61535085564	-312423635629	458868789237	87929221701
170672758566	38764535856	-353594550451	-426781297660
190129283214	-16661605203	326911515143	-141050021129
622296000543	110064546146	279280132741	-178841214672
164935065523	150113758271	-104476022689	-152086655955
64896414169	338399913129	-448538782974	-124926029262
35073244690	-97947594128	283582442608	337418737381
29096501310	271493406321	144605442799	-327769583390
-39267095782	-527046202228	-121357908561	530467454559
303967369490	-284179023506	558696280742	111574982716
321007896070	122544087831	274614721319	-279592934050
340731422122	-79583087940	104947372430	331649781087